Indices

Questions 1–19

Question 1. Without using a calculator, evaluate:

a) $4^{\frac{1}{2}}$
b) $9^{-\frac{1}{2}}$
c) $7^3 \times 7^{-1}$
d) $2 \times 3^3 \times 3^{-2}$
e) $3^5 \div 3^{-1}$
f) $(64^{\frac{1}{3}} \times 16^{\frac{1}{4}})^{-1}$
g) $(\frac{1}{16})^{-\frac{1}{2}}$
h) $(\frac{27}{8})^{-\frac{1}{3}}$
i) $\frac{64^{\frac{2}{3}}}{8^{\frac{5}{3}}}$

Question 2. Express the following as a single power:

a) $x\sqrt{x}$
b) $x^2\sqrt{x}$
c) $\frac{x}{\sqrt{x}}$
d) $\frac{x^2}{\sqrt{x}}$
e) $(\sqrt{x})^3$
f) $(\sqrt[3]{x})^2$
g) $\frac{1}{\sqrt{x}}$
h) $\frac{4}{(\sqrt{x})^3}$
i) $\frac{x\sqrt{x}}{\sqrt[3]{x}}$

Question 3. Express the following in the form $\sqrt[a]{x^b}$, x^b, $\frac{1}{\sqrt[a]{x^b}}$ or $\frac{1}{x^b}$ as appropriate, where a and b are integers to be found:

a) $x^{\frac{1}{2}}$
b) $x^{-\frac{1}{2}}$
c) x^{-5}
d) $x^{\frac{2}{3}}$
e) $x^{-\frac{3}{4}}$
f) $\frac{1}{x^{-\frac{3}{5}}}$

Question 4. Simplify each of the following:

a) $(4a^{10})^{\frac{3}{2}}$
b) $(a^3b^2c)^{\frac{1}{3}} \times (4a^4b^2c)^{\frac{1}{2}}$
c) $(16ab^2)^{\frac{1}{2}} \times (27a^6b^3)^{-\frac{1}{3}}$
d) $(20a^8bc) \div (5a^2b^2c^{-1})$
e) $(3a^2b^3c)^2 \div (9a^2b^{-2}c^4)^{\frac{1}{2}}$

Question 5. Simplify each of the following:

a) $\frac{(2a)^3\times(3a^2)^5}{(6a^5)^2}$

b) $\frac{(3a)^3\times(4a^2)^2}{(2a^4)^3}$

c) $\left(\frac{a^{\frac{1}{2}}\times a^{\frac{5}{2}}}{(a^3)^4}\right)^{-1}$

d) $(\frac{32}{a^8})^{\frac{3}{5}}\times(\frac{a^7}{4})$

Question 6. Simplify each of the following:

a) $\sqrt{x^3\times x^5}$

b) $\sqrt{x^{\frac{3}{5}}\times x^{\frac{1}{5}}}$

c) $\sqrt{49x^6y^4}\times(27x^3y^9)^{-\frac{1}{3}}$

d) $\sqrt{16x^4y^6}\times(9x^2y^3)^{-\frac{1}{2}}$

Question 7. Give each of the following as sums of terms in the form ax^n:

a) $\sqrt{x}(2x+1)$
b) $(\sqrt{x}+1)(\sqrt{x}-2)$
c) $\frac{x+1}{\sqrt{x}}$
d) $\frac{x^2-1}{\sqrt{x}}$
e) $\frac{4}{x}+\frac{1}{\sqrt{9x}}$
f) $\frac{\sqrt{x}-3}{x\sqrt{x}}$

Question 8. A right-angled triangle has perpendicular sides $7a^{\frac{1}{2}}$ and $24a^{\frac{1}{2}}$. Find

a) the length of the hypotenuse side.
b) the area of the triangle.

Question 9. Solve each of the following:

a) $x^{\frac{1}{2}} = 2$
b) $x^{-\frac{1}{2}} = 3$
c) $x^{\frac{1}{3}} = \frac{1}{2}$
d) $x^{-\frac{1}{2}} = 11$
e) $x^{-\frac{3}{2}} = \frac{8}{27}$
f) $2x^{-\frac{1}{3}} = 3$
g) $x^{-\frac{2}{3}} = 2\frac{7}{9}$

Question 10. Mo was asked to find the solutions to

$$x^{\frac{7}{4}} - 2187 = 0.$$

His solution is shown below.

$$x^{\frac{7}{4}} = 2187$$
$$x^{\frac{1}{4}} = \sqrt[7]{2187} = \pm 3$$
$$x = 3^4 \text{ or } (-3)^4$$
$$x = 81 \text{ or } -81$$

Find his mistake(s) and write a correct solution.

Question 11. Write each of these in the form 8^k:

a) $\sqrt{8}$
b) $8^{10} \times 8^{-12}$
c) $\frac{1}{64}$
d) $\sqrt[3]{64}$
e) $\frac{2}{\sqrt{8}}$
f) $\frac{2^6}{\sqrt{8}}$
g) $\sqrt[3]{16} \times (8^3)^{\frac{1}{9}}$
h) $2 \times \sqrt{64} \times \sqrt[3]{16}$

Question 12. Solve each of the following equations:

a) $2^x \times 2^{\frac{1}{2}} = 2$
b) $5^x \times \sqrt{5} = \frac{1}{25}$
c) $3^{x+1} = 9^x$
d) $4^{2x} = \sqrt{2}$
e) $2^{3x-1} = 32$
f) $2^{3x+2} = 4^{x+5}$
g) $9^{2x-1} = \frac{1}{\sqrt{3}}$
h) $\frac{9^x}{3^{x+2}} = 27$
i) $3^{x^2-x} = 729$

Question 13. Simplify each of the following into a single term (selecting the smaller base):

a) $4^x + 4^x + 4^x + 4^x$

b) $2^{2x} + 4^x$

c) $3 \times 2^{3x} + 5 \times 8^x$

d) $5 \times 16^{2x} - 3 \times 2^{8x}$

e) $3^{2x+1} + 3^{2x}$

f) $8 \times 2^x + 3 \times 2^{x+1}$

g) $2^{2x+4} + 4^x$

Question 14. Simplify each of the following so that there are only positive exponents in the numerator and denominator:

a) $\frac{x+x^{-2}}{x^2y+x^{-1}}$

b) $\frac{x^2y^{-1}+x^{-1}}{xy^2+x^{-2}}$

c) $\frac{x^4-x^{-2}yz^3}{x^2+z^{-1}\sqrt{y}}$

d) $\frac{x^5-x^{-1}yz^4}{x^{-3}+z^2y^{-\frac{1}{2}}}$

Question 15. Solve each of the following equations:

a) $4^{2x} \times 3^{2x} = 12$

b) $5^{x} \times 2^{2x} = 20$

c) $3^{2x} \times 9^{x} = 9$

d) $\frac{2^{x}}{2^{\frac{x}{2}} \times 2^{4}} = 1$

Question 16. Find the solutions to $4^{x+2} - 2^{2x+3} - 16 = 0$.

Question 17. Find the value of k such that $\frac{2^{501}-2^{497}}{30} = 64^k$.

Question 18. Solve $\frac{2^x}{32\times 2^{\sqrt{x}}} = 2$.

Question 19. Find one solution to $(x^2 - 4x - 1)^{3x+2} = \frac{1}{4}$.